

Search Platform Architecture

Components

The search platform contains an ingestion service, OpenSearch cluster, query API, and ranking service.

Retrieval

The first stage retrieves candidates using BM25. A second-stage learning-to-rank service may rerank the top 50 candidates.

Why Two Stages

Candidate retrieval must be inexpensive at query volume. Reranking is more expensive and is applied to a smaller candidate set.

Failure Behavior

If the ranking service times out, the API returns the BM25 ordering rather than failing the whole request.